Kurtosys Studio Monorepo — User Guide

Repository: `ksys-studio-template`

One folder is enough. This monorepo covers **Fund Centre** and **product-page widgets**, plus the nested widget starter (`ksys-studio-widget-template/`). Duplicate a single root for each client — no sibling template repo. The first decision is **where embeds run** (Kurtosys DXM vs external CMS). That choice factors builds, deploy paths, and docs without forking the app source.

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Step 0 — Choose host mode (start here)

> **Do this first.** Before adding features or filling GUIDs, decide how the widgets will be embedded. Run init and answer the host question (or pass flags).

```
npm run init
# interactive prompts: host mode → client slug → CMS platform (if needed)

# or non-interactive:
npm run init -- --host dxm --client acme
npm run init -- --host cms --client acme --cms-platform pantheon
npm run init -- --host both --client acme --cms-platform wordpress
```

hostMode	Use when	Bundle behaviour	Primary docs / output
`dxm`	Kurtosys DXM / Studio / Elementor pages	IIFE, **React external**	`hosts/dxm/` · `BuildResult/dxm/…`
`cms`	Client WordPress / Pantheon / custom CMS	IIFE, **React bundled** (self-contained)	`hosts/cms/` · integration specs · `BuildResult/cms/…`
`both`	Same widgets delivered to DXM *and* CMS	Builds both variants from one source	Both host folders · dual `BuildResult` trees

> You do **not** need two monorepos. Host mode only changes packaging, deploy path, and documentation. Widget UI/source stays under `apps/`.

Re-run later with `npm run init -- --host cms --force` (or `npm run host:set`) if the delivery model changes.

1. What this repo is

> **Single-folder template.** Copy `ksys-studio-template` once per client. The widget starter lives at `ksys-studio-widget-template/` inside that copy and is what `npm run apps:add` scaffolds from. You do not need a separate `ksys-studio-widget-template` checkout beside it.

A **generic, reusable monorepo** for Fund Centre and product-page embeds. Each app is a Vite package under `apps/`, registered in `studio.config.json`.

> **Required default: `appManager`.** Every client project must keep `ksys-<client>-appManager/`. It loads child product-page scripts by Studio GUID and publishes co-deployed shared CSS (`shared.css`). Do not rename or remove it (`apps:rename` will refuse).

Starter apps:

App id	Mount id	Embed global	Dev port
`appManager` ★	`#appManager`	`window.appManager`	5170
`fundCentre`	`#fundCentre` (+ `#fundCenter` config)	`window.fundCentre`	5178
`pageHeader`	`#pageHeader`	`window.pageHeader`	5173
`overview`	`#overview`	`window.overview`	5174
`performance`	`#performance`	`window.performanceApp` *	5175
`portfolioBreakdown`	`#portfolioBreakdown`	`window.portfolioBreakdown`	5176
`literature`	`#literature`	`window.literature`	5177

> ★ Required for all projects. Build output: `BuildResult/<host>/<env>/appManager/appManager.js + shared.css` only. Do not ship `bp-shared.css`. Fill `childAppGuids` in its `config/env.config.cjs` after deploying each child widget. See `apps/ksys-<client>-appManager/MONOREPO.md`.

> Fund Centre also needs `defaultFundlist` and optional `subadvisedLandingUrl` in its `config/env.config.cjs` (must be `https://` or a same-site path; anything else is ignored) (see `apps/ksys-<client>-fundCentre/MONOREPO.md`).

> * `window.performance` is reserved by the browser; the loader uses `performanceApp` automatically.

2. Prerequisites

- **Node.js** 20+ and **npm**
- **Python 3.10+** for security reports and PDF generation
- Optional: Chrome / Edge for PDF export

> The nested `ksys-studio-widget-template/` ships with this repo. Keep it when you duplicate the folder; `studio.config.json` → `widgetTemplatePath` points at `./ksys-studio-widget-template`.

3. Repository layout

```
ksys-studio-template/
├── USER_GUIDE.html / USER_GUIDE.pdf
├── studio.config.json          ← hostMode + apps registry + widgetTemplatePath
├── ksys-studio-widget-template/ ← in-repo widget starter (used by apps:add)
├── hosts/
│   ├── ACTIVE_HOST_MODE.txt
│   ├── dxm/                    ← DXM checklist & notes
│   └── cms/                    ← CMS embed notes
├── docs/
│   ├── ACCESSIBILITY.md
│   └── integration-specs/      ← one markdown spec per CMS product page
├── scripts/                    ← init, add/rename, build-all
├── tools/                      ← security audit + user-guide PDF
├── BuildResult/
│   ├── dxm/{dev,stg,prd}/
│   └── cms/{dev,stg,prd}/
└── apps/
    ├── ksys-<client>-appManager/   ★ required default (shell + shared.css)
    ├── ksys-<client>-fundCentre/
    └── ksys-<client>-<appId>/
```

4. Initialize a client project

Duplicate **only** `ksys-studio-template`. Fund Centre, product-page apps, and the nested widget starter come along automatically — do not copy a second template folder.

1. Copy the whole monorepo and rename it for the client:

```
cp -R ~/Documents/GitHub/ksys-studio-template ~/Documents/GitHub/acme-studio
cd ~/Documents/GitHub/acme-studio
# confirm the nested starter + fund centre are present:
ls ksys-studio-widget-template/scripts/scaffold-app.sh
ls apps/ksys-client-appManager/config/env.config.cjs   # required
ls apps/ksys-client-fundCentre/config/env.config.cjs
```

2. **Init host mode** (Step 0).
3. If the client slug is not still `client`, rename folders:

```
npm run client:set -- acme
```

(If you already passed `--client acme` to init, still run `client:set` when folders were created as `ksys-client-*`.)

4. `npm run apps:list` — confirm `hostMode` and app folders.
5. Fill each app's `config/env.config.cjs`.

5. DXM vs CMS — how the project is factored

Shared (always)

- One source tree per widget
- A11y defaults, Tailwind `important`, security pins
- App add / rename / client slug tooling

DXM-only factoring

- `HOST_MODE=dxm` → React **externalized**
- Studio GUIDs + Fundpress / applicationManager URLs
- Outputs under `BuildResult/dxm/<env>/`
- Guidance in `hosts/dxm/README.md`

CMS-only factoring

- `HOST_MODE=cms` → React **bundled** (self-contained for WP/Pantheon)
- Integration spec per product page (Confluence-style)
- Outputs under `BuildResult/cms/<env>/`
- Guidance in `hosts/cms/README.md`

`both`

`npm run build:stg` builds each app twice and collects into both `BuildResult/dxm/…` and `BuildResult/cms/…`.

6. Manage apps

Apps are registered in `studio.config.json` and live under `apps/ksys-<clientSlug>-<appId>/`. Use the commands below from the **monorepo root** (not inside an individual app).

```
npm run apps:list
npm run apps:add -- holdingsExposure 5178
npm run apps:rename -- literature materials
npm run client:set -- acme
```

Add a new app for this project

Use this when the product page needs a widget that is not already in the starter set (for example holdings, distributions, key statistics).

1. Pick a camelCase `appId` (mount id / filename stem) and an optional free Vite port.
2. Scaffold and register it:

```
npm run apps:add -- holdingsExposure 5178
# port is optional — if omitted, the next free port from 5173 is used
```

3. Confirm it appears:

```
npm run apps:list
```

4. Open the new folder and fill config:

```
cd apps/ksys-<clientSlug>-holdingsExposure
# edit config/env.config.cjs  (guid, tokens, containerId, embedGlobalName)
npm install
npm run dev
```

5. Build with the rest of the monorepo when ready:

```
npm run build:stg
```

> `apps:add` copies from the nested `ksys-studio-widget-template/`, writes `apps/ksys-<clientSlug>-<appId>/`, sets the Vite port / starter label, and appends the app to `studio.config.json`. After that, treat it like any other starter app.

> If the host page already has a mount node, keep `containerId` / the DXM `id="…"` in sync with your `appId` (or rename deliberately — see below).

Rename an existing app to match the project

Use this when a starter name does not match the client's language (for example rename `literature` → `materials`, or `portfolioBreakdown` → `holdings`).

```
npm run apps:rename -- <oldId> <newId>
# example:
npm run apps:rename -- literature materials
```

> **Do not rename `appManager`.** It is the required default shell for every project. The command refuses renaming to or from `appManager`.

> **Goal:** after a rename, the old name should not remain anywhere that still describes this widget — *inside* the repo or *outside* it on host pages. That keeps mounts, filenames, docs, and deploys aligned.

1. **In the repo (automated).** `apps:rename` rewrites the old name to the new name across the monorepo:
   - App folder (`apps/ksys-<client>-<oldId>/` → `…-<newId>/`)
   - Package name, mount id, filename, embed global, labels inside that app
   - `studio.config.json` registry entry
   - Other in-repo references (READMEs, `docs/integration-specs/`, `hosts/`, etc.)
   - `BuildResult/…/<oldId>/` folders when present

   It then prints what changed and flags any leftover in-repo mentions to review.
2. Confirm locally:

   ```
   npm run apps:list
   # folder is now apps/ksys-<clientSlug>-materials
   # mount / filename / embed global use the new id (e.g. #materials, materials.js)
   ```

3. **Outside the repo (required — manual).** Update every host or doc that still uses the old name to the new name:
   - DXM / Elementor / App Manager mount nodes (`id="literature"` → `id="materials"`)
   - Script / CDN URLs (`literature.js` → `materials.js`)
   - Studio child-app / GUID wiring that points at the old filename
   - WordPress / Pantheon / CMS embeds and tag-manager snippets
   - Confluence / tickets / runbooks that cite the old mount or file

   The command prints this checklist when it finishes — treat it as part of the rename, not optional cleanup.

4. Re-confirm `config/env.config.cjs` GUIDs still match the (renamed) Studio app.

> The rename is only “done” when the old id is gone from both this repo and the live host pages. Leaving one side on the old name causes blank mounts and mismatched uploads.

> `apps:rename` only renames **one widget id**. To change the client prefix on *every* folder (`ksys-client-*` → `ksys-acme-*`), use `npm run client:set -- acme` instead — that is not an app rename. Template instructional files (`USER_GUIDE.html` / `.pdf`) are left unchanged on purpose; regenerate the PDF after you edit the HTML if your handoff copy should list the new starter names.

Client slug vs app id

Goal	Command	Example result
Brand the whole monorepo for a client	`npm run client:set -- acme`	`ksys-client-overview` → `ksys-acme-overview`
Rename one widget to match the page	`npm run apps:rename -- literature materials`	`…-literature` → `…-materials`
Add a brand-new widget	`npm run apps:add -- holdingsExposure`	new `apps/ksys-acme-holdingsExposure/`

Folder convention: `ksys-<clientSlug>-<appId>/`

7. Develop a single app

```
npm run dev -- overview
```

Local `index.html` simulates App Manager + mount node for DXM-style inputs. For CMS work, the same mount `data-input-*` attributes are still the preferred contract.

8. Configure environments

Each app `config/env.config.cjs` : `dev` / `stg` / `prd` (aliases: `development` → `dev` , `staging` → `stg` , `production` | `prod` → `prd`).

- `guid` , `apiToken` , `savedSearchToken`

- `filename`, `containerId`, `embedGlobalName`

> Do not commit production (prd) secrets to public remotes.

9. DXM mount contract

```
<div
  data-ksys-app-template-id="ksys-app-manager"
  data-input-client-code="…"
  data-input-culture="en-US"
  data-input-product-type="mutualFunds"
></div>
<div id="overview"></div>
<script src="…/overview.js"></script>
```

See `hosts/dxm/README.md`.

10. CMS integration specs

For WordPress / Pantheon (and similar), create **one integration spec per product page** — the same idea as your Confluence “Product Page Strategies / All Regions” templates.

1. Copy `docs/integration-specs/_TEMPLATE.md`
2. Rename to the product page (and region if needed)
3. Fill mount selectors, script URLs, cultures, QA checklists

```
cp docs/integration-specs/_TEMPLATE.md \
   docs/integration-specs/acme-global-equity-us.md
```

See `hosts/cms/README.md` for embed snippets.

11. Build for dev / stg / prd

```
# uses studio.config.json hostMode
npm run build:dev
npm run build:stg
npm run build:prd

# force a single host even if mode is "both"
npm run build:stg:dxm
npm run build:stg:cms
```

Artifacts:

```
BuildResult/dxm/stg/overview/overview.js
BuildResult/cms/stg/overview/overview.js
```

> **Clean artifacts only.** Each app build ends with `clean-dist-artifacts`, which deletes scaffolding (`index.html`, `vite.svg`, manifests, embed-loader, etc.) and keeps only the deployable `<appId>.js` (plus `shared.css` from `appManager`). `BuildResult` folders stay equally lean.

> App Manager build example: `BuildResult/dxm/stg/appManager/{appManager.js,shared.css}`. Deploy the JS + CSS together; child widgets ship their own `<appId>.js` only.

> Per-app scripts also exist: `build:dxm:stg`, `build:cms:stg`, etc. Monorepo `build-all` sets `HOST_MODE` and prefers those scripts.

12. Built-in defaults

- **App Manager** — required shell in every project (`apps/ksys-<client>-appManager/`); loads children by GUID, publishes shared CSS, and marks nameless host tracking pixels decorative (Level Access BP-662 safety net)
- **A11y** — WCAG-oriented landmarks, contrast tokens, focus rings, jsx-a11y, `accessibleImageProps` (see `docs/ACCESSIBILITY.md`)
- **Tailwind** — `@import "tailwindcss" important;`
- **Embed global** — `window.<appId>` (not `YourAppEmbed`)
- **Security pins** — vite, rollup, tar, picomatch, postcss

13. Security audit reports

```
pip install -r tools/requirements-audit-report.txt
npm run audit:security
```

HTML/PDF under `reports/`, scoped to app packages and `BuildResult` bundles.

The template follows the Kurtosys Secure Delivery & Engineering Standards Guide v4. Rules for code you add:

- HTML from the CMS or API goes through `safeHtml()`; links and redirects from config or data go through `safeUrl()` (`src/utils/safe-content.ts`).
- Never put the API token in a URL, a log line or browser storage. Send it only as the `x-ksys-token` header to the Kurtosys API host.
- Production builds strip `console.log`, `info` and `debug`. Use `console.warn` / `error` only for messages support needs, with no tokens or personal data.
- Pin dependency versions exactly, use releases at least two weeks old, and run `npm audit --omit=dev` in each app: no high or critical issues may ship.

14. End-to-end checklist

1. Duplicate `ksys-studio-template` → rename for the client (keep `appManager`)
2. `npm run init -- --host dxm|cms|both --client <slug>`
3. `npm run client:set -- <slug>` if folders still use the old slug
4. Keep `appManager` (required). Rename/add other apps as needed — never rename `appManager`
5. Fill each `env.config.cjs`; for App Manager fill `childAppGuids` after child Studio uploads
6. If CMS/both: create integration specs from `_TEMPLATE.md`
7. Implement UI; keep a11y patterns (`accessibleImageProps` — never omit `alt`)
8. `npm run build:stg` → verify `BuildResult/<host>/…`
9. Deploy (Studio upload and/or CMS CDN) — keep required `appManager` on the page
10. Confirm host tag managers mark tracking pixels with `alt=""` (App Manager also guards BP-662-style beacons)
11. Level Access (or equivalent) scan on a sample new product page
12. `npm run audit:security`

15. Troubleshooting

Symptom	Fix
`hostMode is not set` on build	Run `npm run init` first
CMS page has no React / blank widget	Build with `HOST_MODE=cms` (React must be bundled)
DXM double-React / hook errors	Use `HOST_MODE=dxm` (React externalized)
Styles overridden by theme / Elementor	Confirm Tailwind `important` import in `src/index.css`
Subadvised link or redirect does nothing	`subadvisedLandingUrl` must be `https://` or a same-site path; other values are blocked
Disclaimer HTML loses a tag or attribute	It was removed by `safeHtml()` as unsafe (scripts, event handlers, forms, `javascript:` links)
No `console.log` output on stg/prd	Expected: production builds strip it. Debug with `npm run dev`
`apps:add` cannot find template	Confirm `ksys-studio-widget-template/` was kept when you copied the monorepo, and that `widgetTemplatePath` in `studio.config.json` is `./ksys-studio-widget-template`

16. Regenerate this PDF

```
npm run docs:user-guide
# or: python3 tools/build_user_guide_pdf.py
```

Edit `USER_GUIDE.html` (source of truth), then regenerate the PDF.

Kurtosys Studio product-page monorepo · User guide · Duplicate one folder (widget starter included) · then `npm run init` so host mode factors the rest of the project correctly.